**1 Planning (15 marks)**

**Defining the problem (3 marks)**

P1 Vary *v* and measure *d*, or *v* is the independent variable and *d* is the dependent variable [1]

P2 Keep mass constant [1]

P3 Keep the wood constant/keep same type of nails [1]

**Methods of data collection (5 marks)**

M1 Diagram of apparatus showing mass falling onto centre of nail [1]

M2 Change height of falling mass (to change *v*) [1]

M3 Measurement(s) from which *v* can be determined, e.g. measure height fallen; light gate(s) connected to timer/data-logger measuring time, and ticker tape/motion sensor. Do not award stopwatch methods. [1]

M4 Appropriate equation to determine *v* (the velocity of the mass at the instant it hits the nail) [1]

M5 Detail on measuring *d* ; subtract, needle, mark nail, depth gauge [1]

**Method of analysis (2 marks)**

A1 Plot a graph of log *d* against log *v* [1]

A2 *n* = gradient [1]

**Safety considerations (1 mark)**

S1 Precaution linked to falling masses, e.g. keep well away/sand trays [1]

**Additional detail (4 marks)**

D 1/2/3/4 Relevant points might include [4]

1. Method to create a large *d*, e.g. large mass, thin nails, soft wood
2. Use of a guide for falling mass/guide for nail
3. Use of vernier scale to measure *d*
4. Repeat experiment and determine an average
5. Use different part of wood for each test
6. Method to make nail vertical e.g. set square
7. Discussion / preliminary experiment about thin nails going totally into wood
8. lg *d* = *n* lg *v* + lg *k*

**[Total: 15]**

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